

SEC P vs NP log 2026-05-19

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-19 02:40 UTC

SEC P vs NP — daily log 2026-05-19

15 cycles recorded

Decision Tree Depth and Communication Complexity via Lifting

- **Verdict:** INCONCLUSIVE
- **Bridge:** `DECISION_TREE_DEPTH` $\times$ `COMMUNICATION_COMPLEXITY`
- **Recorded:** 2026-05-19 00:11 UTC
- **Entry ID:** eeec3f948dcf

Statement

For any Boolean function f on n variables, if the decision tree depth of f is d , then the deterministic communication complexity $R_{\{1/3\}}(f)$ is $\Omega(d / \log n)$.

Rationale

This conjecture explores the relationship between the depth of a decision tree and the communication complexity of the function, leveraging lifting techniques to establish a lower bound. If true, it would imply that functions with deeper decision trees require more communication, which is a key insight in understanding the interplay between these complexity measures.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1302.4207v1] A composition theorem for decision tree complexity - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [0911.3482v5] Complexity of Networks (reprise) - [0101006v1] On Complexity and Emergence

Empirical Test

- exit code: 124, elapsed: 240.11s

TIMEOUT after 240s

Judge reasoning

The test timed out after 240s (returncode 124) without producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. | next: Reduce the parameter grid (e.g., restrict to $n \in \{8, 16, 24\}$ and sample fewer depths) and/or optimize the deterministic CC computation to fit within the time budget.

Laplacian Apolar Rank Caps Delorme-Poljak Max-CUT SoS-2 Gap

- **Verdict:** INCONCLUSIVE
- **Bridge:** Curto-Fialkow real moment problem / apolarity & Hankel flat-extension theory (real algebraic geometry). The Hankel rank of the Laplacian moment sequence $m_k(G) := \text{tr}(L_G^k)$ is by the Hamburger truncated moment theorem the smallest support size of an atomic measure on $\mathbb{R}_{\geq 0}$ reproducing the moments — equivalently the number of distinct Laplacian eigenvalues (Curto-Fialkow 1996; Lasserre 2009 ‘Moments, Positive Polynomials and their Applications’; Blekherman-Parrilo-Thomas 2012 ‘Semidefinite Optimization and Convex Algebraic Geometry’; Reznick 1992). The invariant uses real-eigenvalue clustering and rank-of-Hankel — non-ring (no $\mathbb{F}_q$ polynomial extension preserves cluster count of real eigenvalues under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. An arXiv search ‘Curto-Fialkow flat extension’ AND (‘Max-Cut’ OR ‘Delorme-Poljak’ OR ‘SoS integrality gap’) returns 0 direct hits with <5 adjacent papers (Lasserre hierarchy convergence proofs, Helton-Nie semidefinite representability), none using Hankel-apolar rank of the Laplacian-moment sequence as an SoS-2 Max-CUT tightness invariant. Distinct from blacklisted Lee-Yang zeros of the cut polynomial (complex roots, not real moment-sequence), Fekete-capacity (transfinite diameter of spectrum), free 4-cumulant excess (Voiculescu NC partition statistics), Kashin L^1/L^2 -flatness (single-eigenvector L^{1/L^2} ratio), and Halász L^2 discrepancy (CDF deviation). $\times$ Degree-2 Sum-of-Squares / Delorme-Poljak integrality gap $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ for the Max-CUT problem on connected k -regular graphs (Goemans-Williamson 1995; Delorme-Poljak 1993; Schoenebeck 2008; Barak-Steurer 2014). The structural form of the Hankel-rank invariant (a count of distinct Laplacian eigenvalues, a property of the GRAPH WIRING — two non-isomorphic graphs with identical Max-CUT value can have very different $h(G)$) shields it from the Razborov-Rudich natural-proofs barrier; using only real-eigenvalue grouping and integer counting it lies outside the algebrization closure of Aaronson-Wigderson.
- **Recorded:** 2026-05-19 00:24 UTC
- **Entry ID:** 378d9a4dead5

Statement

For every connected k -regular simple graph G on $n \in [8, 20]$ vertices with $k \in \{3, 4\}$, let $h(G)$ be the number of distinct nonzero Laplacian eigenvalues of L_G (equivalently the rank of the truncated Hankel matrix $H_d(G)_{\{i,j\}} = \text{tr}(L_G^{\{i+j+1\}})$ for $d \geq n$, the Curto-Fialkow apolar rank of the Laplacian moment sequence). Let $\text{MC}(G)$ be the exact Max-CUT value and $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ be the Delorme-Poljak SoS-2 integrality gap. Then $n \cdot \rho(G) \leq h(G) \cdot \log_2(n+1)$; a single instance with $n \cdot \rho(G) > 2 \cdot h(G) \cdot \log_2(n+1)$ refutes the conjecture.

Rationale

Apolarity / Curto-Fialkow theory identifies the Hankel rank of a moment sequence with the minimum number of atoms in a representing measure — a canonical real-algebraic-geometry obstruction. Graphs whose Laplacian spectra collapse onto few atoms (distance-regular, SRG, complete bipartite) tend to admit tight Delorme-Poljak SoS-2 certificates because the symmetry forces the optimal pseudoexpectation onto a low-dimensional invariant subspace. The conjecture promotes the folk ‘symmetric spectra = tight SoS-2’ heuristic to a quantitative apolar bound, suggesting Hankel rank as the moment-matrix property P sought by the SOS_DEGREE focus.

Novelty

- Judge: NOVEL over 7 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [2404.18142v1] Revisiting Majumdar-Ghosh spin chain model and Max-cut problem using variational quantum algorithms - [2411.11721v4] Eigenvalues of the Neumann magnetic Laplacian in

- **Recorded:** 2026-05-19 00:32 UTC
- **Entry ID:** fd57783d2ed4

Statement

Let P be a read-twice oblivious BP with width w and 0/1-valued layer transition matrices T_p^b in $\{0,1\}^{w \times w}$ for layers p in $[4n]$ and bits b in $\{0,1\}$, with each variable v read at exactly two positions $p_1(v) < p_2(v)$. Define the bit-flip operator $D_p := T_p^1 - T_p^0$ in $Z^{w \times w}$ and the cross-read Kronecker sum $K(P) := \sum \text{over all } 2n \text{ variables } v \text{ of } D_{p_1(v)} \otimes D_{p_2(v)}$ in $Z^{w^2 \times w^2}$, and let $\rho(P) := \text{rank}_Q(K(P))$. Conjecture: (a) $\rho(P) \leq w(P)^2 \leq s(P)^2$ for every read-twice oblivious BP P (trivial dimension cap), and (b) for every read-twice oblivious BP P computing IP_2 under the adversarial order above, $\rho(P) \geq 2^n / 4$. Equivalently, any read-twice oblivious BP for IP_2 has size at least $2^{\{n/2\}/2}$; a single (P, IP_2) pair with $\rho(P) < 2^{n/4}$ falsifies (b).

Rationale

In Mulmuley-Sohoni GCT the orbit-closure of a polynomial is probed by $GL(w)$ -equivariant tangent directions; the bit-flip operators D_p are exactly the layer-wise infinitesimal generators of the BP's representation variety, and their cross-read Kronecker product encodes the plethystic coupling that READ-ONCE BPs cannot create. For IP_2 under adversarial order, each variable v forces the two reads at $p_1(v), p_2(v)$ to jointly implement an AND-XOR coupling: when bits are flipped the resulting $D_{p_1(v)} \otimes D_{p_2(v)}$ term must contribute a fresh rank-1 block to $K(P)$, and the n distinct variables produce $2n$ rank-1 blocks whose supports cannot algebraically cancel without enlarging the width. The fooling-set / matrix-rank machinery is non-ring and structural on the BP wiring (not the truth table), shielding the bridge from both Razborov-Rudich natural proofs and Aaronson-Wigderson algebrization.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1601.02745v1] Basic Reasoning with Tensor Product Representations - [1511.07136v1] Identity Testing and Lower Bounds for Read- k Oblivious Algebraic Branching Programs - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [2107.09834v4] Communication lower bounds for nested bilinear algorithms via rank expansion of Kronecker products

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 91, in <module>
    trial = run_trial(seed)
            ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 61, in run_trial
    rho_ip2 = compute_rho(n)
              ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 28, in compute_rho
    P_IP2 = build_read_twice_bp(n)
            ~~~~~^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f8c519ad.py", line 23, in build_read_twice_bp
    D_p.append([[T_p_1[i][j] - T_p_0[i][j] for j in range(w)] for i in range(w)])
                ~~~~~^
```

TypeError: 'int' object is not subscriptable

Judge reasoning

The test crashed with a `TypeError` in `build_read_twice_bp` before producing any rho data, so neither the support nor falsification condition could be evaluated. | next: Fix the BP construction bug (ensure `T_p_0` and `T_p_1` are 2D $w \times w$ matrices, not ints) and rerun the pre-registered protocol for n in $\{2,3,4,5\}$.

SoS Cone-Gram Stable Rank Caps AC^0 Depth-d Size for PARITY

- **Verdict:** INCONCLUSIVE
- **Bridge:** Sum-of-Squares hierarchy / Lasserre stable-rank geometry of structural Gram matrices on cone-indicator features (Lasserre 2001; Parrilo 2003; Barak-Steurer 2014; Schur 1923 Hadamard-product PSD theory; Rudelson-Vershynin 2007 stable rank). The invariant $\psi(C) := \sum_i \lambda_i(K(C)) / \lambda_{\max}(K(C))$ — the Schatten-1/Schatten- ∞ stable rank of the gate-cone-indicator Gram matrix $K(C)[g,h] = |\text{cone}(g) \cap \text{cone}(h)|/n$ — is the SoS-degree-2 second-moment / pseudoexpectation functional of the structural feature ensemble $\{1_{\{\text{cone}(g)\}}/\sqrt{n} \mid g \in V(C)\}$ viewed as a Lasserre relaxation of the gate-incidence polytope. It is a non-ring, real-spectral Schatten-ratio (no F_q polynomial extension preserves $\sum \lambda / \lambda_{\max}$ under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. STRUCTURAL on the circuit DAG (two AC^0 circuits with the same Boolean function can have very different cone Gram spectra), so it sidesteps the Razborov-Rudich barrier that killed the prior ‘Lasserre Moment-Rank Gap of Accept Set’ attempt (which operated on truth-table accept sets). $\text{Sym}(n)$ -equivariant under input permutations ($K(C)$ transforms by simultaneous permutation, preserving spectrum), fulfilling the focus’s natural-group-action requirement. An arXiv search ‘Lasserre stable rank’ AND (‘ AC^0 ’ OR ‘PARITY lower bound’ OR ‘switching lemma’) returns 0 direct hits with <5 adjacent papers (all on SoS-Max-Cut / SoS-CSP integrality gaps, never on gate-cone Gram spectra of AC^0 circuits). $\times AC^0$ circuit size $s(C)$ and depth d for the explicit n -bit PARITY function, in the Håstad 1986 switching-lemma / Razborov-Smolensky / Beame 1994 regime, accessed through the depth- d $2^{\Theta(n \{1/(d-1)\})}$ lower bound (open: tight constants and a non-restriction-based proof). In the Cook-Reckhow-Krajíček bounded-arithmetic framework, an AC^0 PARITY size lower bound is the canonical $V^0 \sqsubset V^0(\oplus)$ separation stepping stone.
- **Recorded:** 2026-05-19 00:39 UTC
- **Entry ID:** 63e0e3f43696

Statement

For every AC^0 circuit C of depth $d \geq 2$ on n input variables that computes PARITY_n exactly, the gate-cone-indicator Gram matrix $K(C) \in \mathbb{R}^{|V| \times |V|}$ with $K[g,h] := |\text{cone}(g) \cap \text{cone}(h)|/n$ satisfies $\psi(C) := (\sum_i \lambda_i(K(C))) / \lambda_{\max}(K(C)) \leq C_0 \cdot (\log_2 s(C))^{d-1}$, equivalently $\log_2 s(C) \geq \psi(C)^{1/(d-1)/C_0}$, for an absolute constant $C_0 \leq 4$. The bound is non-vacuous because PARITY requires every input in $\text{cone}(\text{output})$, forcing $\psi(C)$ to scale at least as $n^{(d-2)/(d-1)}$ on any depth- d AC^0 realization, matching Håstad’s lower-bound shape without invoking random restrictions. A single AC^0 circuit C computing PARITY with $\psi(C)^{1/(d-1)} > 4 \cdot \log_2 s(C)$ falsifies the conjecture.

Rationale

The cone Gram matrix $K(C)$ is the natural SoS-degree-2 Lasserre moment-matrix of the structural feature ensemble $\{1_{\{\text{cone}(g)\}}\}_g$, and its stable rank $\psi(C)$ is a Schur-Horn-type combinatorial second-moment measuring how many ‘effective independent input directions’ the wiring of C must explore — exactly the resource that Håstad’s switching lemma shows PARITY forces large via random restrictions, but here computed deterministically from the DAG alone. Because $\text{cone}(g)$ depends only on the wiring (not the gate functions or truth table), two AC^0 circuits with identical Boolean function can have very different $\psi(C)$, giving a Razborov-Rudich-safe structural functional within the SoS hierarchy. The bound’s exponent $(d-1)$ mirrors Håstad’s tight $2^{\Theta(n \{1/(d-1)\})}$

shape, so if it holds even loosely it yields a restriction-free route to AC^0 PARITY lower bounds and a candidate ψ for the focus's polynomial-time natural-invariant target.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 151, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 108, in run_trial
    cones = [reverse_bfs(g, i) for i in range(n)]
              ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7c7fd2fe.py", line 28, in reverse_bfs
    if not visited[neighbor]:
        ~~~~~^~~~~~
```

TypeError: list indices must be integers or slices, not tuple

Judge reasoning

The test crashed with a `TypeError` in `reverse_bfs` (list indexed by a tuple) before producing any $r(C)$ values, so neither the support nor falsification condition could be evaluated. | next: Fix the cone-construction bug in `reverse_bfs` (ensure neighbor is an integer node id, not a (node, edge) tuple) and rerun the 540-instance sweep.

Talagrand L1 Influence Spread of Clause-Falsification Polynomial Lower-Bounds Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Talagrand L1-influence-vector / KKL-Friedgut Boolean Fourier analysis — the Talagrand 1994 functional $T(p) := \sum_i \sqrt{\text{Inf}_i(p)}$, with $\text{Inf}_i(p) = \sum_{S \ni i} \hat{p}(S)^2$, applied as a SYN-TACTIC Fourier-analytic invariant of the clause-falsification pseudo-Boolean polynomial $p_F(x) := \#\{C \in F : x \text{ falsifies } C\}$ of a CNF (a property of the clause LIST — multiplicities and polarities — not of the truth table; two unsat 3-CNFs with identical satisfiability function $\equiv \text{FALSE}$ can have very different $T(F)$). Distinct from blacklisted Mansour L1 mass $\sum_S |\hat{p}(S)|$ (a coefficient-indexed absolute-value sum), Bonami 4/2 ratio $\|p\|_4 / \|p\|_2$ (a hypercontractive norm ratio), Bourgain NSM $\sum_S (1 - \rho^{|S|}) \hat{p}(S)^2$ (a level-weighted L^2 mass), hypercontractive term-pair noise log-ratio (a Rényi-2-to-4 kernel statistic), and cross-side Fourier symmetric-difference mass (a truth-table communication-matrix Fourier object that hit NATURAL_PROOFS) — Talagrand L1 is the unique coordinate-indexed $\sum_i \sqrt{(\sum_{S \ni i} \hat{p}(S)^2)}$ functional with an OUTER $\sqrt{}$ over per-variable L^2 influence buckets. The functional uses $\sqrt{}$ on integer-valued partial sums and a per-coordinate sum that does NOT extend to a polynomial F_q lift (square-root of an L^2 influence has no polynomial-ring surrogate), Aaronson-Wigderson algebrization-safe; structural shielding from Razborov-Rudich since $T(F)$ is not a poly-time-computable property of the truth table of F (which is $\equiv \text{FALSE}$). ArXiv ‘Talagrand L1 influence sum’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers (KKL/Friedgut juntas, never on syntactic clause-falsification polynomials). × Tree-like Resolution / DPLL refutation size $t(F)$ for unsatisfiable 3-CNFs F in the Ben-Sasson-Wigderson 2001 / Beame-Karp-Pitassi-Saks 2002 / Krajíček 1995 bounded-arithmetic regime where $\neg F$ is

Σ^b_0 and $\log_2 t(F)$ lower-bounds V^0 -witnessing complexity; the canonical Frege/Extended-Frege depth stepping stone via Pudlák-Buss feasible interpolation.

- **Recorded:** 2026-05-19 00:46 UTC
- **Entry ID:** 9e12285a05e1

Statement

For every unsatisfiable 3-CNF F on $n \leq 40$ variables with m clauses, let $p_F : \{0,1\}^n \rightarrow \mathbb{Z}_{\geq 0}$ count falsified clauses, expand p_F in the Walsh basis (only subsets S with $|S| \leq 3$ are non-zero), and define $T(F) := \sum_{i=1}^n \sqrt{(\sum_{S \ni i, 1 \leq |S| \leq 3} \hat{p}_F(S)^2)}$. Then $\log_2(1 + t(F)) \geq 0.10 \cdot T(F) / \sqrt{m}$, where $t(F)$ is the DPLL refutation size (backtrack count) under the unit-propagation + first-unassigned variable order. A single F violating this inequality (i.e. $\log_2(1+t^*(F)) < 0.10 \cdot T(F)/\sqrt{m}$) refutes the conjecture.

Rationale

Clause-falsification influence $\text{Inf}_i(p_F)$ measures how strongly variable i ‘pivots’ the count of broken clauses, and Talagrand’s L^1 -spread functional $\sum_i \sqrt{\text{Inf}_i}$ interpolates between the variance $\sum_i \text{Inf}_i (= I(p_F))$, the total clause-sensitivity) and the maximum influence KKL bound, capturing per-variable contention without collapsing to a coefficient sum (Mansour) or a hypercontractive norm ratio (Bonami). Resolution / DPLL must spend $\geq \Omega(\text{width})$ steps to refute, and Ben-Sasson-Wigderson’s width-size tradeoff together with Talagrand’s variance-spread inequality suggest the clause-list-side $T(F)$ functional should track refutation width, hence $\log t^*$. The bridge is structural (not a truth-table property), uses non-polynomial operations ($\sqrt{\cdot}, |\cdot|$), and lies outside both the Razborov-Rudich constructivity regime and the Aaronson-Wigderson algebrization closure.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 175, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 132, in run_trial
    if not is_unsatisfiable(cnf):
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in is_unsatisfiable
    n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.replac
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_972a393d.py", line 33, in <genexpr>
    n = max(int(literal[2:]) if literal.startswith('x') else int(literal[3:]) for literal in cnf.replac
            ~~~~~
```

ValueError: invalid literal for int() with base 10: ''

Judge reasoning

The test crashed with a ValueError during CNF parsing (is_unsatisfiable) before any instances were evaluated, so no violation count or Pearson correlation was produced. The pre-registered support condition cannot be assessed. | next: Fix the CNF literal parser in is_unsatisfiable (handle the negation prefix and empty tokens correctly) and rerun the pre-registered sweep over (n, α) with 30 seeds.

Border Rank Lower Bound for Communication Complexity of Disjointness

- **Verdict:** INCONCLUSIVE
- **Bridge:** ALGEBRAIC_GEOMETRY (border rank of tensors) $\times$ COMMUNICATION_COMPLEXITY
- **Recorded:** 2026-05-19 01:09 UTC
- **Entry ID:** d76acad80bde

Statement

The randomized communication complexity of the DISJ_n matrix is $\Omega(\log(\text{border_rank}(M)))$, where $\text{border_rank}(M)$ is the border rank of the communication matrix over the real numbers.

Rationale

The border rank of a tensor captures its asymptotic complexity, and higher border ranks imply greater information-theoretic requirements for communication. This conjecture links algebraic geometry's tensor rank framework to communication complexity, leveraging the known $\Omega(n)$ lower bound for DISJ_n as a baseline.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:1608.07486] A $2n^2 - \log(n) - 1$ lower bound for the border rank of matrix multiplication - [RNX025] A $2n^2 - \log_2(n) - 1$ lower bound for the border rank of matrix multiplication

Empirical Test

- exit code: 0, elapsed: 0.20s

```
_value': 24, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 24, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 18, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 15, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 12, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 4, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 22, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 20, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 6, 'instances_tested': 1, 'conjec
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 35, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 13, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 38, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 16, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'communication_complexity', 'metric_value': 14, 'instances_tested': 1, 'conje
RESULT: SUPPORTED mean=19.5 std=10.465021102065045 support_fraction=1.0
```

Judge reasoning

The test output only reports a single 'communication_complexity' integer per trial with no border_rank values shown, so the pre-registered criterion (comparing $R(\text{DISJ}_n)$ to $\log_2(\text{border_rank_mean})$) cannot be verified; even if computed, the bound $\log(\text{border_rank}) \leq n$ is trivially satisfied by $R(\text{DISJ}_n) = \Theta(n)$, making the 'SUPPORTED' result vacuous. | next: Re-run with explicit border_rank values logged per n, and compare against a nontrivial benchmark function where border rank is provably super-pol

Communication Complexity Lower Bound for Disjointness Implies Monotone Circuit Size Lower Bound

- **Verdict:** INCONCLUSIVE
- **Bridge:** COMMUNICATION_COMPLEXITY × MONOTONE_CIRCUIT_LOWER_BOUNDS
- **Recorded:** 2026-05-19 01:23 UTC
- **Entry ID:** bf58b88b218f

Statement

For the disjointness function on n -bit inputs, if the deterministic communication complexity is $\Omega(n)$, then every monotone circuit computing it requires size $\Omega(2^{\{n/2\}})$

Rationale

The Karchmer-Wigderson game links communication protocols to circuit depth, but monotone circuits require stronger lower bounds. Disjointness has known $\Omega(n)$ communication complexity, and its monotone circuit size lower bound remains open. A direct connection would unify these complexity measures via combinatorial geometry of protocols.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:2506.12420] Deterministic Lifting Theorems for One-Way Number-on-Forehead Communication - [3406325.3451069] Lower bounds for monotone arithmetic circuits via communication complexity

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 89, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 67, in run_trial
    D = compute_communication_complexity()
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 46, in compute_communication_complexity
    protocol_tree_search(inputs, 0)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

[Previous line repeated 1 more time]

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 44, in protocol_tree_search
    protocol_tree_search(right_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search
    protocol_tree_search(left_partition, depth + 1)
```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_01beaf18.py", line 43, in protocol_tree_search(left_partition, depth + 1)
[Previous line repeated 989 more times]
RecursionError: maximum recursion depth exceeded

Judge reasoning

The test crashed with a RecursionError in protocol_tree_search before producing any data, so neither the communication complexity D nor the monotone circuit size S was measured. With no metrics computed and the critic challenging, the pre-registered support condition cannot be evaluated. | next: Rewrite protocol_tree_search iteratively (or cap depth at $2n$ and prune by rectangle monochromaticity) so $D(\text{DISJ}_4)$ and $S(\text{DISJ}_4)$ can actually be computed before re-running the 10-seed trial.

Cubical First-Betti Linearly Lower-Bounds DNF-MCSP

- **Verdict:** INCONCLUSIVE
- **Bridge:** Cubical combinatorial topology — the cubical subcube complex $K_f := \{\sigma \subseteq \{0,1\}^n : \sigma \text{ is a subcube and } \sigma \subseteq f^{-1}(1)\}$ with its boundary maps over F_2 , $\beta_1(K_f)$ computed as $\text{nullity}(\partial_1) - \text{rank}(\partial_2)$ (Kaczynski-Mischaikow-Mrozek 2004 ‘Computational Homology’; Edelsbrunner-Letscher-Zomorodian 2002; Carlsson 2009 ‘Topology and data’). A non-ring, F_2 -rank-of-cubical-boundary functional: ranks of integer cellular boundaries do NOT extend to a polynomial F_q lift of the Boolean ring (the cubical boundary involves signed face inclusion not preserved by polynomial ring lifts), so the bridge is Aaronson-Wigderson algebrization-safe. The invariant is RELATIONAL (a categorical functor $(\text{Bool}^n \rightarrow \text{Top} \rightarrow \text{Vect}\{F_2\})$ applied to the subcube cover poset), not a single truth-table number — sidestepping trap (j). An arXiv search ‘cubical homology’ AND (‘MCSP’ OR ‘minimum DNF’ OR ‘meta-complexity’) returns 0 direct hits with <5 adjacent papers (all on TDA of biological data or persistent-homology classifiers, never on DNF-MCSP). Distinct from blacklisted Mahler-measure (single complex-analytic truth-table number), FCA antichain width (Galois-lattice width on assignments $\times$ literals), treewidth of PI compatibility graph (graph-minor invariant), and independence-complex Euler $\tilde{\chi}$ (Stanley-Reisner alternating sum). $\times$ DNF-MCSP / minimum DNF size $\sigma_{\text{DNF}}(f)$ for $f: \{0,1\}^n \rightarrow \{0,1\}$ in the Quine-McCluskey / Masek 1979 / Umans 1999 / Allender-Hellerstein-McCabe-Pitassi-Saks / Hirahara-Oliveira meta-complexity regime, where σ_{DNF} is the optimum of a prime-implicant set-cover and the canonical NP-hard meta-complexity object whose hardness fuels Hirahara’s worst-case-to-average-case program. Sparsity escape from Razborov-Rudich: ‘ $\sigma_{\text{DNF}}(f) \leq s$ ’ is rare among random f (only $\sim 2^{\{O(s \log n)\}}$ functions admit it, of $2^{\{2n\}}$ total), so any lower bound $\sigma_{\text{DNF}} \geq \varphi(f)$ inherits sparsity from the cap side and does not violate Razborov-Rudich largeness.
- **Recorded:** 2026-05-19 01:33 UTC
- **Entry ID:** 9020e29f840c

Statement

Let $K_f = \{\sigma \subseteq \{0,1\}^n : \sigma \text{ is a subcube and } \sigma \subseteq f^{-1}(1)\}$ be the cubical subcube complex of $f^{-1}(1)$, and $\beta_1 := \beta_1(K_f, F_2)$ its first F_2 -Betti number. Then for every Boolean $f: \{0,1\}^n \rightarrow \{0,1\}$ satisfying $\beta_1(K_f) \geq 1$, the minimum DNF size obeys $\sigma_{\text{DNF}}(f) \geq 3 \cdot \beta_1(K_f)$. One counterexample (a single f with $\sigma_{\text{DNF}}(f) < 3 \cdot \beta_1(K_f)$ and $\beta_1(K_f) \geq 1$) refutes the conjecture.

Rationale

By the nerve theorem applied to any DNF cover $F = T_1 \vee \dots \vee T_s$ of f (each T_i a subcube $\subseteq f^{-1}(1)$), the nerve N_F is homotopy equivalent to K_f , giving $\beta_1(N_F) = \beta_1$. Trivially N_F has s vertices and any 1-cycle uses ≥ 3 of them, yielding only $\sigma_{\text{DNF}} \geq \sqrt{(2\beta_1)}$; the conjecture asserts the much stronger LINEAR bound — each independent topological hole costs 3 fresh terms in any

optimal cover because subcubes encircling a hole are ‘rigid’ (cannot be shared across distinct holes without collapsing the cycle). This is a clean nerve-rigidity strengthening pioneered by no prior MCSP work.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 105, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 84, in run_trial
    beta_1, sigma_DNF = dnf_size(f, n)
                        ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 25, in dnf_size
    beta_1 = compute_beta_1(K_f)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_80da2c5c.py", line 30, in compute_beta_1
    n = len(K_f[0])
        ~~~~^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in compute_beta_1 (empty K_f list) before any instances were evaluated, so no data was produced to assess the pre-registered support or falsification condition.
| next: Fix compute_beta_1 to handle the empty subcube complex case ($\beta_1 = 0$ when K_f is empty) and rerun the 270-triple sweep.

Fourier Spectral Entropy of Clause-Falsification Polynomial Bounds Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier-analytic information theory — Shannon entropy of the normalized Walsh-Hadamard spectrum of a multilinear pseudo-Boolean polynomial (Friedgut-Kalai 1996 FEI conjecture; O’Donnell 2014 ‘Analysis of Boolean Functions’ Ch. 9; Wan-Wang-Yu 2014 on FEI for low-degree functions; Chakraborty-Kulkarni-Lokam-Saurabh 2016 ‘Upper bounds on Fourier entropy’). The functional $H_F(F) := -\sum_S q_F(S) \cdot \log_2 q_F(S)$, where $q_F(S) := \hat{p}_F(S)^2 / \|p_F\|^2$, is the Shannon entropy of the SPECTRAL probability distribution viewed as a measure on the Boolean character lattice 2^1 . It is distinct from Mansour $L^1 \sum |\hat{p}(S)|$ (linear in $|\hat{p}|$, no log), Bourgain NSM $\sum_S (1 - \rho^{|S|}) \hat{p}(S)^2$ (level-indexed L^2 weight, not entropy), Bonami $4/2$ ratio $\|p\|_4 / \|p\|_2$ (norm ratio), Talagrand $\sum_i \sqrt{\text{Inf}_i}$ (per-coordinate L^2 -root), term-pair noise log-ratio (Rényi-2-to-4 hypercontractive kernel), and cross-side symdiff mass (truth-table CC-matrix Fourier). It uses log of a normalized ratio of squares, so it admits no polynomial $F \rightarrow q$ extension (Shannon entropy of a probability distribution is not preserved under polynomial-ring lifts of the Boolean ring), making the bridge Aaronson-Wigderson algebrization-safe. Razborov-Rudich-safe because $H_F(F)$ is a property of the CLAUSE LIST (multiplicities,

¹_n

polarities), so two unsat 3-CNFs F_1, F_2 with identical satisfiability function (both $\equiv \text{FALSE}$) can have very different H_F . An arXiv search ‘Fourier entropy’ AND (‘DPLL’ OR ‘tree-Resolution’ OR ‘CNF refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers (all on FEI for ± 1 Boolean f or random-DNF entropy, never on the clause-falsification polynomial of an unsat CNF). $\times$ Tree-like Resolution / DPLL refutation size $t(F)$ for unsatisfiable 3-CNFs F in the Chvátal-Szemerédi 1988 / Ben-Sasson-Wigderson 2001 / Beame-Karp-Pitassi-Saks 2002 / Krajíček 1995 bounded-arithmetic regime, where $\neg F$ is Σ^b_0 and $\log_2 t(F)$ lower-bounds V^0 -witnessing complexity; the canonical Frege/Extended-Frege depth stepping stone via Pudlák-Buss feasible interpolation.

- **Recorded:** 2026-05-19 01:40 UTC
- **Entry ID:** 363fda00c17f

Statement

Let F be an unsatisfiable 3-CNF on n variables with m clauses. Define $p_F : \{-1, +1\}^n \rightarrow \mathbb{R}$ by $p_F(x) := \sum_{C \in F} \mathbb{1}[x \text{ falsifies } C]$; its Walsh-Hadamard spectrum $\hat{p}_F(S)$ is supported on $|S| \leq 3$ and admits the closed form $\hat{p}_F(\emptyset) = m/8$, $\hat{p}_F(\{i\}) = -\sum_{C \ni i} \varepsilon_i^C / 8$, $\hat{p}_F(\{i, j\}) = \sum_{C \supseteq \{i, j\}} \varepsilon_i^C \varepsilon_j^C / 8$, $\hat{p}_F(\{i, j, k\}) = -\sum_{C = \{i, j, k\}} \varepsilon_i^C \varepsilon_j^C \varepsilon_k^C / 8$ where $\varepsilon_i^C \in \{\pm 1\}$ is the polarity of variable i in clause C . Let $q_F(S) := \hat{p}_F(S)^2 / \|p_F\|^2$ and $H_F(F) := -\sum_{\{S : |S| \leq 3\}} q_F(S) \cdot \log_2 q_F(S)$. Conjecture: there exists a universal constant $c > 0$ such that for every unsat 3-CNF F at clause-to-variable ratio $\alpha \in [4.2, 5.0]$, $\log_2 t(F) \geq c \cdot n \cdot H_F(F) / \log_2(1 + n + n^2 + n^3)$. One counterexample (an unsat 3-CNF whose ratio $\log_2 t(F) \cdot \log_2(1 + n + n^2 + n^3) / (n \cdot H_F(F))$ is arbitrarily small) refutes it.

Rationale

The spectral entropy H_F measures how DELOCALIZED the Walsh-Hadamard energy of the clause-falsification polynomial is across degree- ≤ 3 character modes; intuitively, a CNF whose Fourier mass concentrates on a few low-degree characters admits a short tree-Resolution refutation (a few decisions kill the dominant modes), while a CNF whose Fourier mass spreads broadly forces DPLL to branch in many directions. The Friedgut-Kalai FEI heuristic — entropy is bounded by total influence — suggests an inverse linear relationship between spectral concentration and combinatorial complexity, and here we are normalizing entropy by the log of the number of degree- ≤ 3 characters to get an n -scaled functional. The bridge is genuinely under-explored: spectral entropy of the syntactic clause polynomial has never been used as a DPLL lower-bound certificate.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 139, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 111, in run_trial
    spectrum = walsh_hadamard_transform(cnf)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6e131d60.py", line 59, in walsh_hadamard_transform
    spectrum = [0] * (2 ** n_vars)
                ^~~~~~
```

OverflowError: cannot fit 'int' into an index-sized integer

Judge reasoning

The test crashed with an OverflowError when attempting to allocate a 2^n -sized array for the Walsh-Hadamard transform, so no $R(F)$ values or Pearson correlations were produced. Without data, neither the support nor the falsification condition can be evaluated. | next: Rewrite `walsh_hadamard_transform` to exploit the closed-form spectrum (support only on $|S| \leq 3$, $O(n^3)$ coefficients) rather than allocating a 2^n array, then rerun the pre-registered protocol.

2-Adic Cokernel Type of Sign Matrix Caps Randomized CC

- **Verdict:** INCONCLUSIVE
- **Bridge:** Hall algebra of finite abelian 2-groups / Cohen-Lenstra heuristics on integer-matrix cokernels — the partition $\lambda_2(M) \vdash v_2(|\det M|)$ encoding the 2-Sylow structure of $\text{cok}(M) = \mathbb{Z}^n / M \cdot \mathbb{Z}^n$ as a finite abelian 2-group (Hall 1959; Macdonald 1995 ‘Symmetric Functions and Hall Polynomials’ Ch. II ‘The Hall algebra’; Cohen-Lenstra 1984 ‘Heuristics on class groups’; Friedman-Washington 1989 random matrix cokernels; Wood 2017 ‘Random integral matrices’). The invariant is computed via Smith normal form mod 2^N in $O(n^3 \cdot \log n)$ pure-Python big-int Gaussian elimination, sidestepping any Coxeter/Hecke machinery. An arXiv search ‘Cohen-Lenstra’ AND (‘communication complexity’ OR ‘sign rank’ OR ‘discrepancy’) returns 0 direct hits with <5 adjacent papers (all on lattice-based cryptography or chip-firing critical groups, never on CC matrices). Distinct from blacklisted 2-Sylow rank of Tseitin critical group $K(G)$ (an invariant of a charged GRAPH, not a sign matrix) and from Specht-isotype / cycle-type / RSK / hook-length attempts (which projected via S_n -characters or partition statistics, never via the Hall-algebra 2-adic cokernel type of an integer matrix). × Two-party randomized communication complexity $R_{\{1/3\}}(M)$ for $\{\pm 1\}$ -sign matrices $M \in \{\pm 1\}^{n \times n}$, in the Yao 1979 / Razborov 1992 DISJ regime, accessed via the Linial-Shraibman γ_2 / Klartag-Regev corruption lower bound and the Forster sign-rank route to UPP^{cc} . The invariant $\lambda_2(M)$ is STRUCTURAL on the sign assignment: it is preserved under the full CC symmetry group (row/column permutation + signed-diagonal pre/post conjugation $D_1 \cdot M \cdot D_2$ with $D_i \in \{\pm 1\}^n$ diagonal), since these act as $\mathbb{Z}$ -module automorphisms of $\mathbb{Z}^n$ and leave the cokernel isomorphism class fixed — but it is NOT a poly-time-computable property of the underlying Boolean function (two CC matrices with identical function up to relabeling can have differing λ_2 after lift to $\mathbb{Z}$), shielding from Razborov-Rudich.
- **Recorded:** 2026-05-19 01:51 UTC
- **Entry ID:** 7227efbf367

Statement

For every sign matrix $M \in \{\pm 1\}^{n \times n}$, let $v_2(|\det M|)$ be the 2-adic valuation of the determinant computed over $\mathbb{Z}$, and define $\rho(M) := v_2(|\det M|) / n \in [0, (\log_2 n)/2]$. We conjecture there exists an absolute constant $c \in (0, 4)$ such that for all $n \geq 4$ and all $M \in \{\pm 1\}^{n \times n}$ with $\gamma_2(M) \leq g$ (the Linial-Shraibman γ_2 -norm, equivalent to $R_{\{1/3\}}(M)$ up to log factors): $\rho(M) \leq c \cdot (\log_2 g) / (\log_2 n)$. One sign matrix M with $\rho(M) \cdot \log_2 n > c \cdot \log_2 \gamma_2(M)$ for $c = 4$ falsifies the conjecture.

Rationale

Sign matrices with arithmetic structure that forces $|\det M|$ to absorb a large 2-power (Sylvester-Hadamard, IP_k , Paley) are precisely those where the $\mathbb{Z}$ -module $\text{cok}(M)$ is far from the Cohen-Lenstra random model — and these are also the canonical hard CC functions (IP_k achieves $R = \Omega(k)$, Hadamard achieves $\gamma_2 = \Theta(\sqrt{n})$). Conversely, low-CC matrices admit small monochromatic-rectangle covers, which when lifted to $\mathbb{Z}$ force $\text{cok}(M)$ to have small 2-Sylow rank because the cover provides a near-rank-1 integer decomposition. The Hall-algebra 2-Sylow type encodes finer structure than rank alone (it tracks the elementary-divisor partition), so it can detect structured matrices that fool spectral / Frobenius-norm CC bounds.

Novelty

- Judge: NOVEL over 9 arXiv hits

Top hits consulted: - [1703.01572v1] The Smith normal form of a specialized Giambelli-type matrix
- [2211.09736v2] Equidistributions of Sign Patterns of the Liouville Function and Normal Numbers
- [1602.00166v2] Smith Normal Form in Combinatorics - [1211.0398v1] Extending valuations to formal completions - [1503.07648v2] Sign rank versus VC dimension

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbb8938a.py", line 151, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbb8938a.py", line 122, in run_trial
    det = abs(det_mod_2k(M, n * math.ceil(math.log2(n))))
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbb8938a.py", line 94, in det_mod_2k
    factor = (A[j][i] * pow(A[i][i], -1, 2**k)) % (2**k)
              ~~~~~
```

ValueError: base is not invertible for the given modulus

Judge reasoning

The test crashed with a ValueError in det_mod_2k (non-invertible base modulo 2^k) before any of the 630 instances produced data, so neither the support nor falsification condition can be evaluated. The critic also challenged due to the crash. | next: Replace the modular inverse-based Gaussian elimination with a direct integer determinant (e.g., sympy.Matrix.det or fraction-free Bareiss algorithm) and re-run the 630-instance sweep.

vdW 3-AP Density of MAJ-Lifted NW Designs Tracks Overlap k

- **Verdict:** INCONCLUSIVE
- **Bridge:** Arithmetic Ramsey theory — quantitative van der Waerden / Behrend AP-density bounds (Behrend 1946; Szemerédi 1975; Bloom-Sisask 2020 ‘Breaking the logarithmic barrier in Roth’s theorem’; Kelley-Meka 2023). The 3-AP density functional ρ_3 on Boolean sequences is a comparison/counting statistic (no F_q polynomial extension preserves monochromatic-AP counts under polynomial ring lifts), Aaronson-Wigderson algebrization-safe. Distinct from Chang spectrum / Plünnecke / additive-energy / Sidon attempts: those measure Fourier-spectrum or sumset-cardinality on truth-table or wire-label objects, while ρ_3 counts monochromatic 3-APs in the OUTPUT POSITION sequence of a MAJ-lifted design. ArXiv ‘van der Waerden’ AND (‘Nisan-Wigderson’ OR ‘NW design’ OR ‘combinatorial design pseudorandom generator’) returns 0 direct hits and <5 adjacent papers (Behrend-style cap-set constructions; never on NW designs). $\times$ Nisan-Wigderson PRG combinatorial-design overlap parameter k controlling seed-blow-up 2^k (Nisan-Wigderson 1994; Impagliazzo-Wigderson 1997; Klivans-van Melkebeek 2002; Kabanets-Impagliazzo 2004), in the BPP=P-under-hardness regime. The invariant is STRUCTURAL on $D = (S_1, \dots, S_m)$ (two designs supporting the same Boolean PRG function can yield very different ρ_3), shielding from Razborov-Rudich natural proofs; the MAJ-lift breaks F_2 ring structure since MAJ has no $o(\sqrt{l})$ -degree polynomial approximation (Razborov-Smolensky), shielding from algebrization.
- **Recorded:** 2026-05-19 02:06 UTC
- **Entry ID:** 4bcf619e2ecb

Statement

Let $D = (S_1, \dots, S_m)$ be an (l, k) -NW design on $[n]$ with $|S_i| = l$ and $\max_{\{i < j\}} |S_i \cap S_j| \leq k$ ($k \geq 1, l \geq 4$). Define the MAJ-lift $\chi_D : \{0, 1\}^n \rightarrow \{0, 1\}^m$ by $\chi_D(z)_i = \text{MAJ}(z|_{S_i})$ (lexicographic tie-break = 0). Let $\rho_3(D) := \mathbb{E}_{\{z \sim U(\{0, 1\}^n)\}} [\#\{(a, d) : 1 \leq a, a+2d \leq m, d \geq 1, \chi_D(z)_a = \chi_D(z)_{a+d} = \chi_D(z)_{a+2d}\}] / N_3(m)$ where $N_3(m)$ is the total number of 3-APs in $[m]$. Conjecture: there exist universal constants $0 < c \leq C < \infty$ (independent of n, l, k, m) such that for every such NW design D , $c \cdot (k/l)^{1/2} \leq \rho_3(D) - 1/4 \leq C \cdot (k/l)^{1/2}$. A single design with $\rho_3(D)$ outside this band falsifies it.

Rationale

NW PRG fooling power degrades like 2^k in the overlap parameter, but k is a global combinatorial property of the design that should manifest in second-order output statistics. Monochromatic 3-AP density is the simplest non-trivial AP statistic and probes 3-wise correlation across position triples $(i, i+d, i+2d)$ — exactly the structure that Behrend/Bloom-Sisask measure. If the conjecture holds, ρ_3 gives a black-box, sub-quadratic-time structural certificate of design overlap, which feeds the Kabanets-Impagliazzo hardness-vs-randomness program by lower-bounding the seed-length penalty of any design substrate from output statistics alone.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c15756de.py", line 79, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c15756de.py", line 54, in run_trial
    chi_D = [MAJ(z, s) for z in range(2**n) for s in S]
            ^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c15756de.py", line 30, in MAJ
    count = sum(1 for x in z if x in S)
            ^^^^^^^^^^^^^^^^^^^^^^^^^
```

TypeError: 'int' object is not iterable

Judge reasoning

The test crashed with a `TypeError` in the `MAJ` function (iterating over an `int` instead of a bit vector) before producing any data, so neither the Pearson correlation nor the R-ratio criterion could be evaluated. | next: Fix the `MAJ` implementation to iterate over the bits of z restricted to S (e.g., represent z as a tuple/list of bits or use bitmask operations) and rerun the pre-registered 30-seed sweep.

Lorentzian Defect of Cut Polynomial Lower-Bounds Max-CUT SoS-2 Gap

- **Verdict:** INCONCLUSIVE
- **Bridge:** Lorentzian polynomials / Brändén-Huh log-concavity and the matroid Hodge-Riemann relations of Adiprasito-Huh-Katz — the normalized log-concavity condition $\tilde{a}_{j-1} \cdot \tilde{a}_{j+1} \leq \tilde{a}_j^2$ with $\tilde{a}_j := a_j / C(m, j)$ characterizing when a homogenized univariate coefficient vector lies in the closure of the Lorentzian cone; the defect $LD := \max_j$

$\max(0, \log(\tilde{a}_{j-1} \cdot \tilde{a}_{j+1} / \tilde{a}_j^2))$ is a sup-of-log-ratios functional that uses only integer counts, binomial normalisation, and log of ratios, with no F_q polynomial extension (binomial normalisation and log are not preserved under polynomial ring lifts), so the bridge is Aaronson–Wigderson algebrization-safe. Distinct from blacklisted Lee–Yang zero cluster angle (complex roots of the same polynomial), Kashin L^1 -flatness (single-eigenvector L^1/L^2 ratio), Curto–Fialkow Hankel-apolar rank (real moment problem on L_G spectrum), free 4-cumulant excess (Voiculescu NC partitions on squared singular values), Fekete capacity (transfinite diameter of Laplacian spectrum), hypercontractive 4/2 ratio (norm ratio of the truth table), and Schur–Horn majorization gap (NATURAL_PROOFS-rejected). An arXiv search ‘Lorentzian polynomial’ AND (‘Max-Cut’ OR ‘Delorme–Poljak’ OR ‘SoS integrality gap’) returns 0 direct hits with <5 adjacent papers (Anari–Liu–Oveis–Gharan–Vinzant on log-concavity for matroid-base sampling, never on SoS-2 integrality gaps). × Degree-2 Sum-of-Squares / Delorme–Poljak Max-CUT integrality gap $\rho(G) := n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ for connected 3-regular graphs G , in the Goemans–Williamson 1995 / Delorme–Poljak 1993 / Schoenebeck 2008 / Barak–Steurer 2014 SOS_HIERARCHY regime; the canonical stepping stone toward unconditional SoS-degree- d Max-CUT lower bounds via matroidal obstructions on degree-2 pseudoexpectation moment matrices.

- **Recorded:** 2026-05-19 02:13 UTC
- **Entry ID:** dad52f056827

Statement

Let G be a connected 3-regular graph on n vertices with $m = 3n/2$ edges and cut-size distribution $a_j(G) := \#\{S \subseteq V(G) : |\delta_G(S)| = j\}$, and let $LD(G) := \max\{0, \max_{\{1 \leq j \leq m-1, a_{j-1}a_{j+1} > 0\}} \log((\tilde{a}_{j-1} \cdot \tilde{a}_{j+1}) / \tilde{a}_j^2)\}$ with $\tilde{a}_j := a_j / C(m, j)$ be the normalized Lorentzian defect of the cut-generating polynomial $P_G(z) = \sum_j a_j z^j$. Then for every such G , $LD(G) \geq 0.05 \cdot \rho(G)$, where $\rho(G) = n \cdot \lambda_{\max}(L_G) / (4 \cdot \text{MC}(G)) - 1$ is the Delorme–Poljak SoS-2 integrality gap. Equivalently, any connected 3-regular G whose cut-polynomial coefficient sequence is normalized log-concave (Lorentzian, $LD=0$) must have $\rho(G) = 0$ (SoS-2 tight).

Rationale

The Brändén–Huh Lorentzian property of a homogeneous polynomial is exactly the matroidal Hodge–Riemann signature condition; if $P_G(z, w)$ (homogenized) were Lorentzian, the symmetric bilinear form on $(R^{\wedge} V, J_G)$ induced by the cut-direction Gram would inherit a one-positive-eigenvalue signature compatible with the Laplacian’s spectral gap, forcing the SoS-2 pseudoexpectation cone to align with the integer-cut cone. Conversely the Delorme–Poljak gap measures the angle between the leading Laplacian eigenvector and the cut polytope’s max-cut face, an angle that combinatorially obstructs the Newton-inequality regularity of (a_j) . Bridging matroid log-concavity to the SoS-2 obstruction is essentially unexplored: it bypasses the NATURAL_PROOFS trap that killed Schur–Horn (the invariant is structural on the graph, not on a truth table) and the ALGEBRIZATION trap that closes ring-only constructions (binomial normalisation and log of ratios admit no F_q polynomial surrogate).

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:2407.19905] The Bidirected Cut Relaxation for Steiner Tree has Integrality Gap Smaller Than 2 - [LIPIcs.SWAT.2024.23] A Logarithmic Integrality Gap for Generalizations of Quasi-Bipartite Instances of Directed Steiner Tree

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b1bc8fd9.py", line 103, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b1bc8fd9.py", line 71, in run_trial
    lambda_max = laplacian_eigenvalues(n, m)[0]
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b1bc8fd9.py", line 42, in laplacian_eig
    return [eigenvalue for eigenvalue in numpy.linalg.eigvalsh(L) if eigenvalue >= 0]
           ^^^^^^
```

NameError: name 'numpy' is not defined

Judge reasoning

The test crashed with a NameError ('numpy' not defined) before any of the 210 instances were evaluated, so no data on the $LD(G) \geq 0.05 \cdot p(G)$ inequality was produced. Per Rule 2, a crashed test yields INCONCLUSIVE. | next: Fix the import (add 'import numpy' or use the already-imported alias) in laplacian_eigenvalues and rerun the full 210-instance sweep.

Level-1 Fourier Mass Fraction of Term Family Bounds Monotone CLIQUE DNF

- **Verdict:** INCONCLUSIVE
- **Bridge:** Fourier-analytic hypercontractivity / KKL low-level concentration on syntactic term-incidence empirical measures $\times$ Monotone DNF leaf size $L(F)$ for the k -CLIQUE indicator F^*_v on K_v with $k = \lfloor \sqrt{v} \rfloor$, $n = v(v-1)/2$ edge variables (Razborov 1985 / Andreev / Alon-Boppana monotone regime)
- **Recorded:** 2026-05-19 02:22 UTC
- **Entry ID:** 9ec4f1b18412

Statement

For a monotone DNF $F = \bigvee_{i=1}^s T_i$ over n variables, view the term-support indicators $\{1\{T_i\}\}_{i=1}^s \subset \{0,1\}^n$ as an empirical sample and let $\hat{\mu}_F(S) := (1/s) \sum_i (-1)^{|S \cap T_i|}$ be the Walsh-Fourier coefficient of the empirical measure at $S \subseteq [n]$. Define the level-1 low-frequency fraction $\lambda(F) := (\sum_{|S|=1} \hat{\mu}_F(S)^2) / (\sum_{1 \leq |S| \leq 2} \hat{\mu}_F(S)^2)$ and the complexity measure $\mu(F) := -\log_2 \max(\lambda(F), 1/n)$. Conjecture: (a) for every monotone DNF F of leaf size $L(F) = \sum_i |T_i|$, $\mu(F) \leq 2 \cdot \log_2(L(F)+2) + 4$; (b) for the k -CLIQUE indicator F^*_v on K_v with $k = \lfloor \sqrt{v} \rfloor$, $\mu(F^*_v) \geq (1/8) \cdot v$ for all $v \geq 4$. Equivalently, $\log_2 L(F^*_v) \geq \Omega(v) = \Omega(\sqrt{v})$, a monotone CLIQUE leaf-size lower bound for $k = \sqrt{v}$.

Rationale

By KKL/Friedgut, a Boolean-valued empirical measure with level-1 mass dominating its low-level spectrum behaves like a junta of few variables — exactly the regime of small DNFs whose terms touch few variables and whose Fourier signal sits on singletons. The k -CLIQUE indicator's term family is the $(v \text{ choose } k)$ edge-sets of k -cliques, an S_v -equivariant design that uniformizes level-1 character sums and pushes ≤ 2 -level mass onto edge-pair correlations, suppressing λ to $O(1/v)$. The measure is SYNTACTIC on the term list (two monotone DNFs computing the same Boolean function can have very different μ), so it is shielded from Razborov-Rudich, and uses $\log_2$ of an $L^{2/L-2}$ fraction over level-1 and level-2 characters — a non-ring functional with no F_q polynomial lift (Aaronson-Wigderson algebrization-safe).

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 196.97s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 94, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 50, in run_trial
    mu_F = complexity_measure(level_1_low_frequency_fraction(F))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 28, in level_1_low_f
    sum_level_1 = sum(walsh_fourier_coefficient(F, {j+1 for j in range(n) if (i >> j) & 1})**2 for i in r
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 28, in <genexpr>
    sum_level_1 = sum(walsh_fourier_coefficient(F, {j+1 for j in range(n) if (i >> j) & 1})**2 for i in r
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8b489d7d.py", line 23, in walsh_fourier
    sum_term += F[i] * term
                ^^^
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in walsh_fourier_coefficient before producing any RESULT line, so neither ensemble (A) nor (B) was evaluated. Per the rules, a crashed test yields INCONCLUSIVE. | next: Fix the indexing bug (likely a 0-based vs 1-based variable indexing mismatch between F and the subset construction {j+1 ...}) and re-run all 30 seeds plus $v \in \{4..8\}$.

Dilworth Width of Gate-Cone Poset Lower-Bounds AC^0 PARITY Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Order theory — Dilworth's theorem and König-matching computation of poset width on Boolean-lattice containment orders (Dilworth 1950 'A decomposition theorem for partially ordered sets'; Mirsky 1971; Fulkerson 1956 'Note on Dilworth's decomposition theorem'; Trotter 1992 'Combinatorics and Partially Ordered Sets'). A non-ring set-inclusion functional: width via Hopcroft-Karp matching on the strict-comparability bipartite graph uses only Boolean subset tests and integer matching, with NO polynomial F_q extension (containment of supports in 2^2 is not preserved under polynomial-ring lifts of the Boolean ring), so the bridge is Aaronson-Wigderson algebrization-safe. $\times AC^0$ depth-d circuit size $s(C)$ for the explicit n-bit PARITY function, in the Håstad 1986 switching-lemma / Razborov-Smolensky / Beame 1994 regime; the canonical $2^{\Theta(n \{1/(d-1)\})}$ target and the $V^0 \sqcap V^0(\oplus)$ bounded-arithmetic stepping stone.
- **Recorded:** 2026-05-19 02:31 UTC
- **Entry ID:** f2533eda1d01

²_n

Statement

Let C be an AC^0 circuit of depth d and size $s = |V(C)|$ with $C \equiv \text{PARITY}_n$, and define the gate-cone poset $P(C)$ on $V(C)$ by $g \sqsubseteq h$ iff $\text{support}(g) \subseteq \text{support}(h)$ where $\text{support}(g) := \{x \in \{0,1\}^n : g(x)=1\}$; let $w(C) := \text{width}(P(C))$ be Dilworth's maximum-antichain size, computed by König matching as $|V(C)| - \mu(B(C))$ on the strict-comparability bipartite graph $B(C)$. Conjecture: there exists an absolute constant $c > 0$ such that $w(C) \geq c \cdot \log_2(s)$ for every AC^0 circuit C computing PARITY_n ; equivalently $s \leq 2^{\{w(C)/c\}}$. A single AC^0 instance $C \equiv \text{PARITY}_n$ with $w(C)/\log_2(s) < c$ for every fixed c refutes it.

Rationale

A pairwise-incomparable family of gate cones is a 'structural fooling set' in the containment lattice — antichains here play the role of rectangle-discrepancy fooling sets that LIFTING transfers from query (decision-tree certificates) to communication, but for SIZE rather than depth, since min chain cover (Dilworth dual) equals the minimum number of monotone monocoloured chains needed to schedule the gate evaluation. The invariant is structural on the circuit wiring — two AC^0 circuits with identical truth table (e.g. one with redundant subsumed gates) can have very different widths, shielding from Razborov–Rudich natural proofs — and the Boolean-lattice subset relation does not algebrize.

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC^0 and TC^0 - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1505.02372v3] On Asymptotic Gate Complexity and Depth of Reversible Circuits With Additional Memory

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 133, in <module>
    trial = run_trial(seed)
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 110, in run_trial
    w_C = hopcroft_karp(build_bipartite_graph(C, assignments))
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 42, in build_bipartite
    G.extend([(g, h) for h in C if set(support(g, assignments)) <= set(support(h, assignments))])
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_00eea011.py", line 37, in support
    return [i for i, bit in enumerate(assignments) if gate[i] == 1]
            ~~~~~
```

TypeError: 'int' object is not subscriptable

Judge reasoning

The test crashed with a `TypeError` in the `support()` function ('int' object is not subscriptable), indicating a bug in how gates are represented or indexed; no instances were evaluated, so neither the support nor falsification condition can be assessed. | next: Fix the gate representation in `support()`: ensure each gate is stored as a truth-table vector (list/array indexable by assignment index) rather than a scalar, then re-run the 360-instance sweep.

Mean Log-Bias of Internal Gates Lower-Bounds AC^0 PARITY Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Information geometry — mean Kullback-Leibler divergence from Bernoulli(1/2) of the internal-gate bias empirical measure (Csiszár 1975 ‘I-divergence geometry of probability distributions’; Amari-Nagaoka 2000 ‘Methods of Information Geometry’; Cover-Thomas 2006). The functional uses log of empirical probabilities and per-gate $\min(p, 1-p)$, with NO polynomial F_q extension (log and min are not preserved under polynomial-ring lifts of the Boolean ring), so the bridge is Aaronson-Wigderson algebrization-safe. An arXiv search ‘mean gate-bias KL divergence’ AND (‘ AC^0 ’ OR ‘switching lemma’ OR ‘PARITY lower bound’) returns 0 direct hits and <5 adjacent papers (all on noise-sensitivity of the OUTPUT function in the Kahn-Kalai-Linial / Bourgain / O’Donnell tradition, never on the empirical measure of INTERNAL gate biases as a structural circuit invariant). Distinct from blacklisted Lasserre Moment-Rank Gap on accept set (truth-table side; NATURAL_PROOFS-rejected), SoS Cone-Gram Stable Rank (Schatten-1/ ∞ spectrum on a Gram matrix), Talagrand L^1 influence spread / Fourier spectral entropy / Mansour L^1 / Bonami 4/2 / Bourgain NSM (all on the OUTPUT polynomial’s Fourier transform, not internal gate biases), and Dilworth width / Gromov hyperbolicity / Mostar / Forman-Ricci (all combinatorial-graph functionals on the DAG, never an information-geometric KL-from-uniform on Bernoulli pseudoexpectations). $\times AC^0$ depth-d circuit size $s(C)$ for the n-bit PARITY function in the Håstad 1986 switching-lemma / Razborov-Smolensky / Beame 1994 regime, targeting the canonical $2^{\Theta(n)}\{1/(d-1)\}$ lower bound and the $V^0 \sqsubseteq V^0(\oplus)$ bounded-arithmetic separation. Gate biases p_g are exactly the SoS-degree-2 pseudoexpectations $\mathbb{E}[1_{\{g(x)=1\}}]$ under the uniform pseudo-distribution, so the invariant is a per-gate average of an SoS-2 functional — a Lasserre-pseudomoment-based circuit invariant in the Barak-Steurer 2014 / Grigoriev SoS hierarchy lineage.
- **Recorded:** 2026-05-19 02:40 UTC
- **Entry ID:** 8f8f4d4c4a54

Statement

Let C be an AC^0 depth-d circuit on n inputs with m internal gates $g_1, \dots, g_m$ (including the output, excluding input literals). Let $p_g := \Pr_{x \sim U_n}[g(x)=1]$ (estimated by $N=\min(2^n, 2^{14})$ random samples) and define $\psi(C) := (1/m) \cdot \sum_g [-\log_2(2 \cdot \max(\min(p_g, 1-p_g), 1/(2m)))]$. CONJECTURE: (a) $\psi(C) \leq \log_2 m + 1$ for every AC^0 circuit C [universal cap]; (b) for every AC^0 depth-d circuit C exactly computing PARITY_n, $\psi(C) \geq 0.2 \cdot n^{\{1/(d-1)\}}$ [PARITY witness]. Combining (a)+(b) yields $\log_2 \text{size}(C) \geq 0.2 \cdot n^{\{1/(d-1)\}} - 1$.

Rationale

Håstad’s switching lemma forces internal AC^0 gates whose downstream cone reaches PARITY’s perfectly balanced output to each carry exponentially small $\min(p_g, 1-p_g) \approx 2^{\{-n\{1/(d-1)\}\}}$; the mean-log-bias condenses this ‘bias bookkeeping’ into a single SoS-pseudoexpectation invariant. The invariant is RELATIONAL on the gate-bias multiset of C (two circuits with identical Boolean function can have very different gate-bias distributions because they wire intermediate gates differently), shielding from Razborov-Rudich natural-proofs and sidestepping the truth-table trap (j) that killed prior Lasserre Moment-Rank attempts. The natural S_n action on input bits permutes inputs but leaves the multiset $\{p_g\}$ fixed (uniform measure is S_n -invariant), giving the required non-trivial group equivariance.

Novelty

- Judge: NOVEL over 10 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC^0 and TC^0
 - [2604.11744v1] KL Divergence Between Gaussians: A Step-by-Step Derivation for the Variational

Autoencoder Objective - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1205.6363v1] What Should Developers Be Aware Of? An Empirical Study on the Directives of API Documentation - [2103.17141v3] Yoneda's lemma for internal higher categories

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_916b5b32.py", line 161, in <module>
    trial = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_916b5b32.py", line 105, in run_trial
    circuit = generate_hastad_circuit(n, d)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_916b5b32.py", line 63, in generate_hastad_circuit
    term &= ('AND', j)
```

TypeError: unsupported operand type(s) for &=: 'bool' and 'tuple'

Judge reasoning

The test crashed with a `TypeError` in `generate_hastad_circuit` (`term &= ('AND', j)` misuses `&` on `bool` and `tuple`) before producing any data, so neither the universal cap nor the PARITY lower bound was empirically evaluated. | next: Fix the circuit construction bug (initialize `term` as a `tuple`/`AST` node rather than a `bool`, e.g., `term = ('AND',)` and `append_children`) and re-run the 30-seed sweep over (n, d) cells for E1/E2/E3.